

WEEK 4
POWER BI EDA

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1. INTRODUCTION

This project is used to analyze stock prices and trading volume using Power BI.

Data containing Date, Open, High, Low, Close and Volume.

2. OBJECTIVES

- To analyze stock data.
- To identify price trends.
- To calculate 20-day and 50-day SMA.
- To analyze trading volume.
- To create an interactive dashboard.
- To use slicers and charts for analysis.

3. DATASET DESCRIPTION

Dataset Name: SHOP_2015-05-21_2025-03-16

Important Columns:

- Date
- Open
- High
- Low
- Close
- Volume

Data Type: Historical stock data.

4. DATA PREPARATION

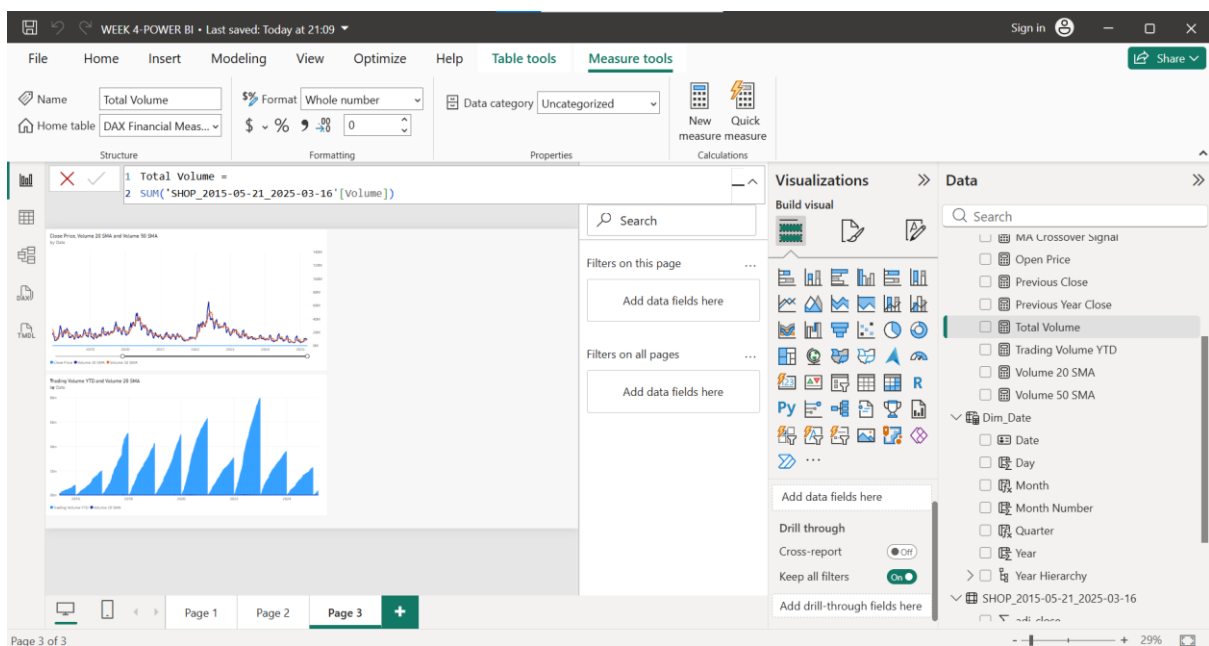
- Data imported into Power BI.
- Date table created.
- Data fields checked.
- DAX measures created.
- Data model prepared.

5. DASHBOARD DESIGN & VISUALIZATIONS

Volume Chart

Used: Date, Total Volume, Volume 20 SMA, Volume 50 SMA and MA Crossover Signal

Purpose: To show trading volume.



File Home Help

Clipboard Paste Get data Excel workbook OneLake catalog SQL Server Enter data Data Warehouse Recent sources Transform Refresh data Queries Manage relationships Relationships New measure column Calculations Calculation group New parameter Parameters Manage roles View as Security Q&A Language setup Linguistic schema Sensitivity Public

```
1 Volume 20 SMA =
2 VAR CurrentDate =
3   MAX(Dim_Date[Date])
4 VAR PeriodDays = 20
5 VAR DateInPeriod =
6   DATESINPERIOD(
7     Dim_Date[Date],
8     CurrentDate,
9     -PeriodDays,
10    DAY
11  )
12 RETURN
13 CALCULATE(
14   AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
15   DateInPeriod
16 )
```

Properties

General

Name: Volume 20 SMA

Home table: DAX Financial Measures (2)

Description: Enter a description

Synonyms: volume 20 SMA, SMA, 20 SMA

Data

Tables Model

Search

DAX Financial Measures (2)

- Close Price
- Close Price LY
- Close Price Previous Year
- Close Price YTD
- Daily Return %
- High Price
- low price
- Open Price
- Previous Close
- Previous Year Close
- Trading Volume YTD
- Volume 20 SMA

Dim_Date

SHOP_2015-05-21_2025-03-16

All tables +

File Home Help

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```
1 Volume 50 SMA =
2 VAR CurrentDate =
3   MAX(Dim_Date[Date])
4 VAR PeriodDays = 50
5 VAR DateInPeriod =
6   DATESINPERIOD(
7     Dim_Date[Date],
8     CurrentDate,
9     -PeriodDays,
10    DAY
11  )
12 RETURN
13 CALCULATE(
14   AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
15   DateInPeriod
16 )
```

Properties

General

Name: Volume 50 SMA

Home table: DAX Financial Measures (2)

Description: Enter a description

Synonyms: volume 50 SMA, SMA, 50 SMA

Data

Tables Model

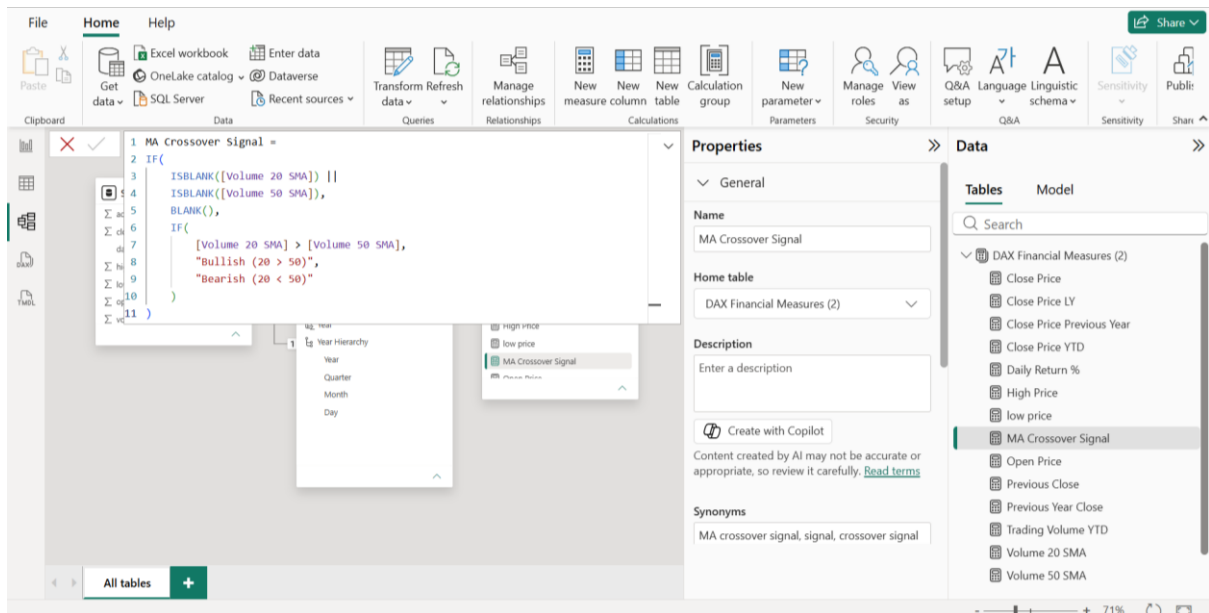
Search

DAX Financial Measures (2)

- Close Price
- Close Price LY
- Close Price Previous Year
- Close Price YTD
- Daily Return %
- High Price
- low price
- Open Price
- Previous Close
- Previous Year Close
- Trading Volume YTD
- Volume 20 SMA
- Volume 50 SMA

Dim_Date

All tables +



Date Slicer

Used: Date

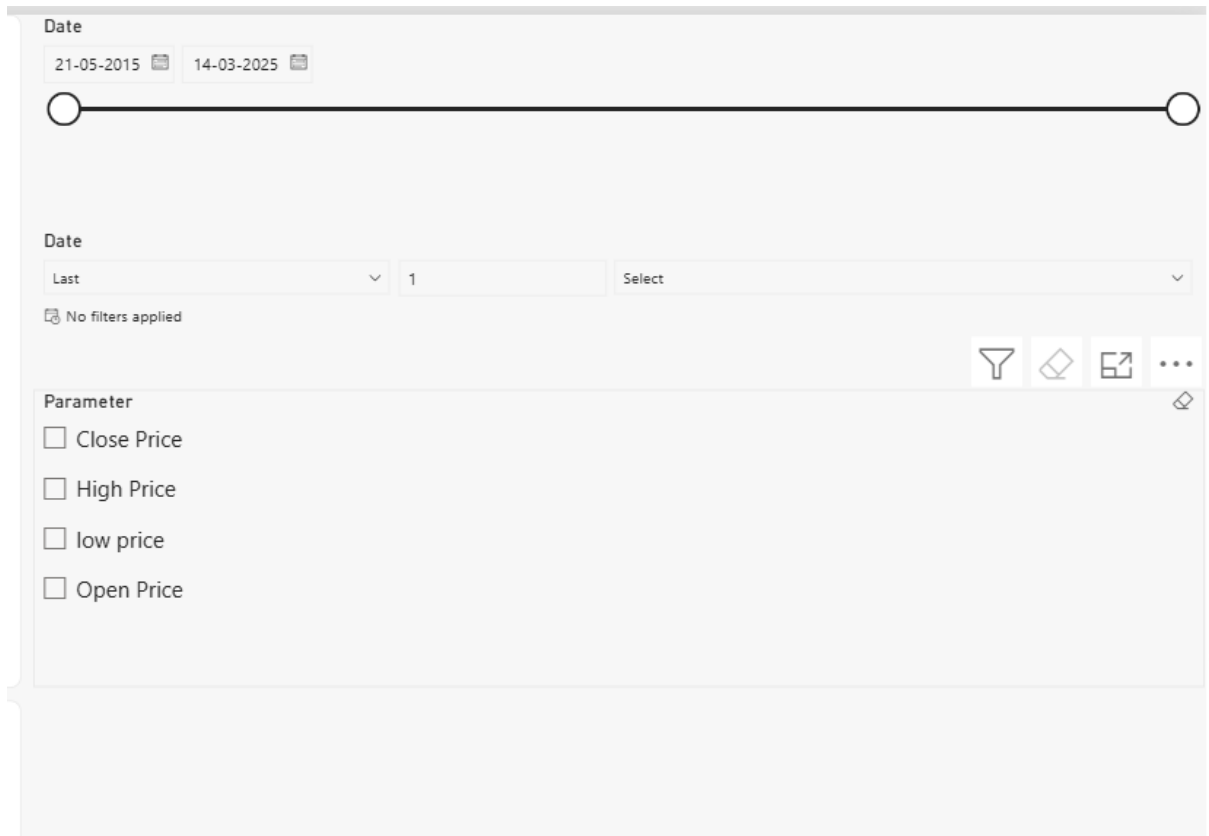
Purpose: To filter dates.

Date	MA Crossover Signal	Volume 20 SMA	Volume 50 SMA
21-05-2015 00:00:00	Bearish (20 < 50)	12,30,39,000.00	12,30,39,000.00
22-05-2015 00:00:00	Bearish (20 < 50)	7,57,25,500.00	7,57,25,500.00
23-05-2015 00:00:00	Bearish (20 < 50)	7,57,25,500.00	7,57,25,500.00
24-05-2015 00:00:00	Bearish (20 < 50)	7,57,25,500.00	7,57,25,500.00
25-05-2015 00:00:00	Bearish (20 < 50)	7,57,25,500.00	7,57,25,500.00
26-05-2015 00:00:00	Bearish (20 < 50)	5,32,17,666.67	5,32,17,666.67
27-05-2015 00:00:00	Bearish (20 < 50)	4,19,07,250.00	4,19,07,250.00
28-05-2015 00:00:00	Bearish (20 < 50)	3,43,25,800.00	3,43,25,800.00
29-05-2015 00:00:00	Bearish (20 < 50)	2,90,30,500.00	2,90,30,500.00
30-05-2015 00:00:00	Bearish (20 < 50)	2,90,30,500.00	2,90,30,500.00
31-05-2015 00:00:00	Bearish (20 < 50)	2,90,30,500.00	2,90,30,500.00
01-06-2015 00:00:00	Bearish (20 < 50)	2,52,50,285.71	2,52,50,285.71
02-06-2015 00:00:00	Bearish (20 < 50)	2,25,68,750.00	2,25,68,750.00
03-06-2015 00:00:00	Bearish (20 < 50)	2,03,97,111.11	2,03,97,111.11
Total	Bearish (20 < 50)	97,86,826.67	96,57,082.86

Parameter Switch

Used: Open, High, Low, Close

Purpose: To select different price values.



Custom Tooltip

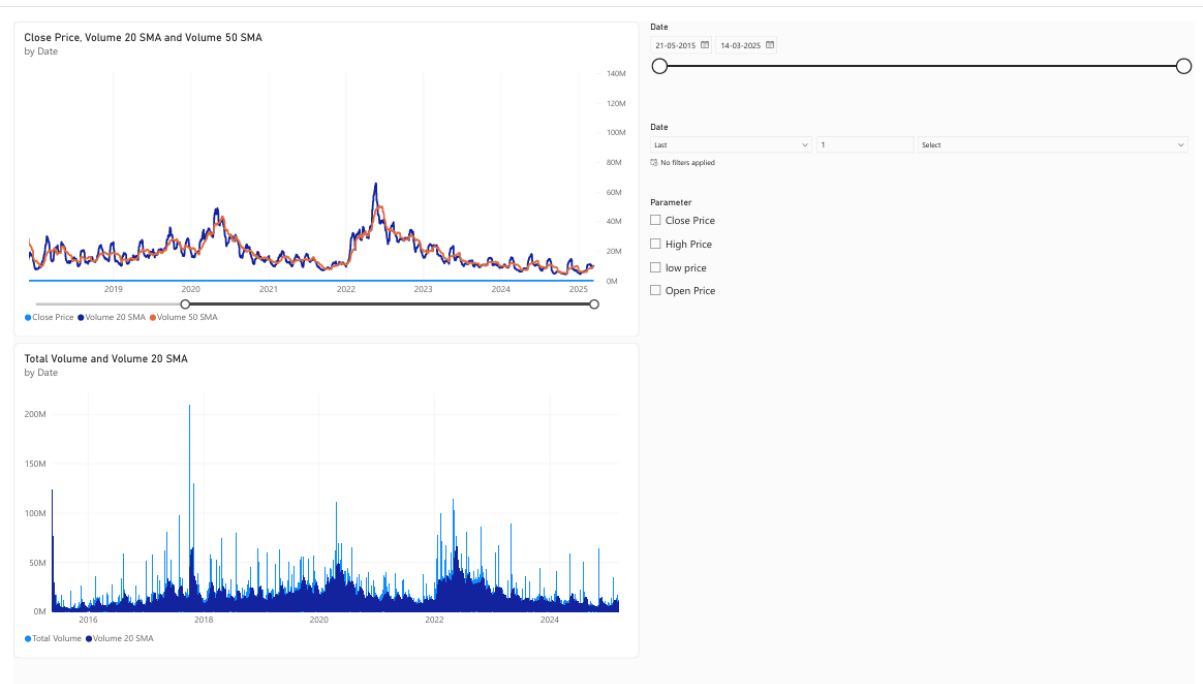
Used: Open, High, Low, Close, Volume, Daily Change %

Purpose: To show details when hovering over the chart.

Open Price	High Price	low price	Close Price	Total Volume
48.46	176.29	1.85	48.46	40912544

Open Price	low price	Close Price	Total Volume	Daily Return %
176.29	1.85	48.46	40912544200	0.00

6. DASHBOARD SCREENSHOT



7. CONCLUSION

This project shows how Power BI can be used to analyze stock market data. The dashboard helps users understand stock price trends, moving averages and trading volume.